

WASP-43b

R-0331

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_260218 · created 2026-07-18T04:38:52+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2461089.777 +/- 0.0024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.161 +/- 0.019
Transit depth (Rp/Rs)^2	2.59 +/- 0.6 [%]
Orbital Inclination (inc)	82.04 +/- 1.73
Airmass coefficient 1 (a1)	0.996 +/- 0.014
Airmass coefficient 2 (a2)	0.004 +/- 0.011
Scatter in the residuals of the lightcurve fit is	1.45 %
Best Comparison Star	#1 - [172, 66]
Optimal Aperture	3.56
Optimal Annulus	11.06
Transit Duration (day)	0.0498 +/- 0.0064

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.161 ± 0.019 vs 0.1594 (deviation 0.08σ)
Duration fit vs published	0.0498 d vs 0.0512 d (deviation 0.23σ)
Mid-time O–C	-0.6 min
Depth SNR	8.5
Residual scatter	1.45 %

Light curve

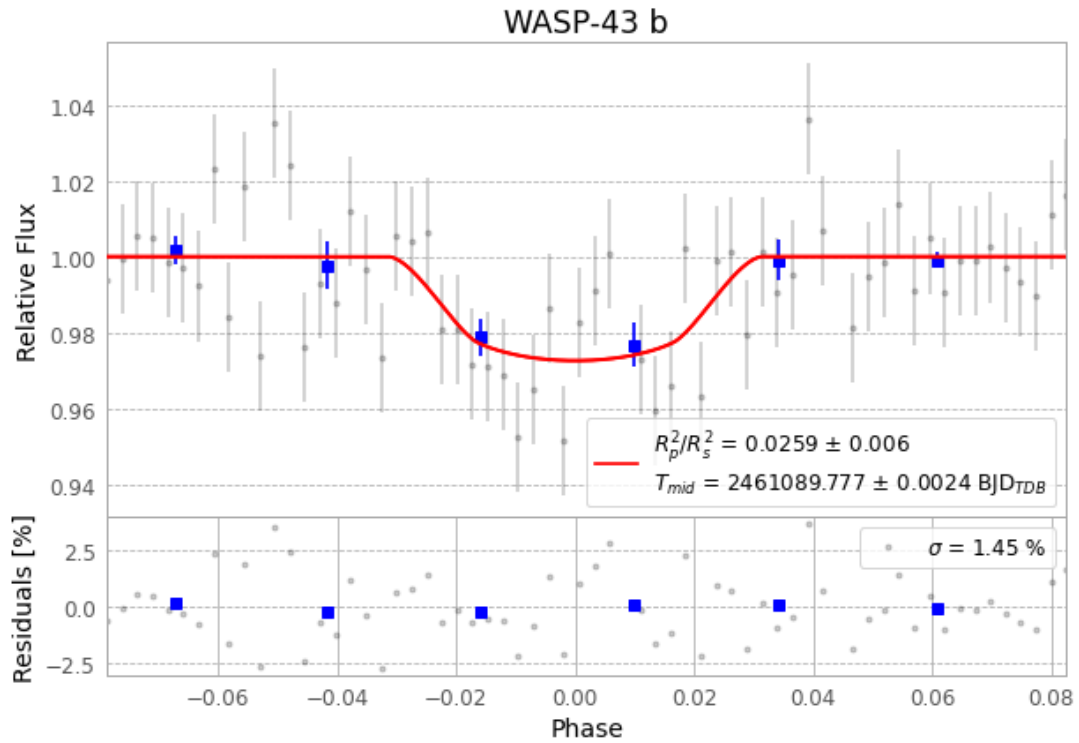


Figure 1 — FinalLightCurve_WASP-43 b_2026-02-17.png (SHA-256 f8c0449bfb544492...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [467, 192], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8b2288aac2470f07...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 600780d

Data provenance

64 FITS frames (42 MB), WASP-43260218045715.FITS → WASP-43260218080615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-260218.log (343ee1a99a33...), FinalLightCurve_WASP-43 b_2026-02-17.png (f8c0449bfb54...), FinalLightCurve_WASP-43 b_2026-02-17.csv (c9016ab3a88a...)

Compute resources

Wall time 225.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 04:38Z.